

ThermalFlow — Safety Conformance Ledger

Technical-documentation evidence, structured to EU AI Act Annex IV
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VERDICT PASS	Checks 12/12 passed failed 0 · errored 0	Code digest (SHA-256) 57430618adf7c6605b70589e4772f22f2272e53572e8d5df2bf9c986aad13dae git n/a
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Scope and legal notice

This ledger is automatically generated conformance EVIDENCE for the ThermalFlow control and safety logic. It is produced by deterministic, seeded, in-process simulation that imports and exercises the production safety modules directly (the files hashed under provenance.code_digest). It is offered as a CONTRIBUTION to a deployer's technical documentation and is structured to the headings of EU AI Act Annex IV (Regulation (EU) 2024/1689) for convenience.

It is NOT, and must not be represented as: (a) a legal determination that ThermalFlow is an 'AI system' or a 'high-risk AI system' under the Act; (b) a statement that Annex IV or any other obligation applies; (c) a conformity assessment, audit, or certification. Whether the Act applies, and under which Annex III category (e.g. safety components in the management of critical infrastructure), is a determination for the deployer's compliance/legal function. The authors are not lawyers and this is not legal advice.

Latencies reported here are CONTROL-LOOP-CYCLE figures under a modelled plant. They do NOT include message-bus transport, container scheduling, or physical-actuator response. Wall-clock latency on real infrastructure MUST be measured separately via a live-stack smoke test before any safety claim is made. The simulation covers control/safety LOGIC; it does not cover the messaging or orchestration layer. No warranty is given.

Provenance

Python	3.12.3
Platform	Linux-6.18.5-x86_64-with-glibc2.39
Git commit	None
Combined code digest	57430618adf7c6605b70589e4772f22f2272e53572e8d5df2bf9c986aad13dae
backend/safety-relay/relay.py	a0c8757e69c39d6264bf8b3446e6b647...
backend/safety-relay/fence.py	3541a7e8bc9a946c8fadca4c8b4f6325...
backend/safety-relay/service.py	49e480411f4c8790688f0bcc05b8db97...
backend/plant-sim/plant_sim.py	b2218fd14752410e70e7fd5a50e0bf74...
backend/plant-sim/service.py	6eab8d9aed400c90c129be5eb2188f7e...
backend/health-monitor/health_monitor.py	083e1a996e8f1eb2ec76bea43d318f30...
backend/mpc/mpc_core.py	fc68c002069725f67404973ca5af91b7...
backend/rom-shadow/rom_model.py	b4080505a580063a9c1be6fcc73de027...

Annex IV coverage map

Best-effort structural mapping of the evidence below to Annex IV headings. Not a completeness claim for the documentation as a whole.

Ref	Annex IV heading (paraphrased)	Evidence in this ledger
§2(c)	System architecture; how components build on/feed each other	Series single-writer topology; relay as sole writer; fencing at the actuator.
§2(g)	Validation and testing procedures and results, including metrics	This entire runner: deterministic conformance checks with measured latencies.
§3	Monitoring, functioning and control; capabilities and limitations	Interlocks (L1/L2/watchdogs), command validation, dead-man, HA fencing.
§4	Appropriateness of the performance metrics	Fallback-latency budgets and alarm-lead metrics stated per check.
§5	Risk management system (Art 9): identified risks and mitigations	Each fault class mapped to its detecting interlock and mitigating fallback.
§9	Post-market monitoring plan	Coil-health drift detector (advisory alarm) with load-bin baselining.

Conformance results

ID	Check	Annex IV	Verdict
CHK-IL-L1-BG	L1 over-temperature, break-glass (single sensor > 38 C)	Annex IV §2(g),§3,§5	PASS
CHK-IL-L1-VOTE	L1 over-temperature, voted (2-of-3 trio > 32 C)	Annex IV §2(g),§3,§5	PASS
CHK-IL-STALE-SENSOR	Sensor watchdog (comms loss > 10 s)	Annex IV §2(g),§3,§5	PASS
CHK-IL-STALE-MPC	MPC watchdog (missed control cycle > 90 s)	Annex IV §2(g),§3,§5	PASS
CHK-IL-INVALID-CMD	Command validation (out-of-bounds / rate violation)	Annex IV §2(g),§3,§5	PASS
CHK-LATCH-DURABLE	Latch durability across restart	Annex IV §3,§5	PASS
CHK-IL-L2-RESIDUAL	L2 model-residual interlock (acute CRAC loss)	Annex IV §2(g),§3,§4,§5	PASS
CHK-DEADMAN	Dead-man revert on lost controller	Annex IV §3,§5	PASS
CHK-FENCE-ZOMBIE	Zombie fencing across failover (no dual-write)	Annex IV §2(c),§3,§5	PASS
CHK-FENCE-RESTART	Restart re-acquisition (fresh sequence accepted)	Annex IV §2(c),§3	PASS
CHK-HEALTH-LEAD	Coil-health early warning precedes safety trip	Annex IV §3,§9 (post-market monitoring)	PASS
CHK-HEALTH-LOADSHIFT	Coil-health load-shift false-alarm immunity	Annex IV §3,§4,§9	PASS

Evidence detail

CHK-IL-L1-BG · L1 over-temperature, break-glass (single sensor > 38 C) **PASS**

Requirement	Any single rack sensor above the hard limit must latch a break-glass trip and command the fast fallback ramp.
Method	Prime ARMED on safe data; inject one sensor at 39.5 C; assert immediate trip.
Measured	armed_before=True, trip_source=L1_single:r4/top=39.5, severity=break_glass, detect_cycles=1
Budget	max_detect_cycles=2
Annex IV	Annex IV §2(g),§3,§5
Note	trip at L1_single:r4/top=39.5

CHK-IL-L1-VOTE · L1 over-temperature, voted (2-of-3 trio > 32 C) **PASS**

Requirement	Two of three sensors in a rack trio above the soft limit must latch a graded trip (single-sensor noise must not).
Method	Prime ARMED; inject a trio with two sensors at ~33 C; assert graded trip.
Measured	trip_source=L1_vote:r2, severity=graded, detect_cycles=1
Budget	max_detect_cycles=2
Annex IV	Annex IV §2(g),§3,§5

CHK-IL-STALE-SENSOR · Sensor watchdog (comms loss > 10 s) **PASS**

Requirement	Loss of sensor data beyond the watchdog horizon must trip (blind => assume unsafe).
Method	Stop the sensor feed while MPC stays fresh; measure seconds to trip.
Measured	trip_source=stale_sensor, trip_s=11
Budget	max_trip_s=12
Annex IV	Annex IV §2(g),§3,§5
Note	break-glass: blind controller assumes unsafe

CHK-IL-STALE-MPC · MPC watchdog (missed control cycle > 90 s) **PASS**

Requirement	A stalled advisory MPC must trip to the safe fallback rather than hold a stale setpoint indefinitely.
Method	Stop MPC proposals while sensors stay fresh; measure seconds to trip.
Measured	trip_source=stale_mpc, trip_s=91
Budget	max_trip_s=95
Annex IV	Annex IV §2(g),§3,§5

CHK-IL-INVALID-CMD · Command validation (out-of-bounds / rate violation) **PASS**

Requirement	A physically implausible MPC proposal (bounds or slew-rate) must be rejected by the relay before it reaches the plant.
Method	Feed an in-bounds proposal that jumps 0.20 in one cycle (> 0.15 slew limit); assert trip.
Measured	trip_source=invalid_command:rate df=+0.20, detect_cycles=1
Budget	max_detect_cycles=2
Annex IV	Annex IV §2(g),§3,§5

CHK-LATCH-DURABLE · Latch durability across restart **PASS**

Requirement	A latched trip must survive an OOM/cold restart and refuse to re-arm without an explicit operator reset.
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Method	Trip + persist; instantiate a fresh relay; restore; assert it boots TRIPPED.
Measured	latch_persisted=True, fresh_relay_armed=True, after_restore=TRIPPED
Budget	—
Annex IV	Annex IV §3,§5

CHK-IL-L2-RESIDUAL · L2 model-residual interlock (acute CRAC loss) **PASS**

Requirement	A divergence between the frozen clean-coil reference and the plant (acute cooling loss) must trip on the residual norm.
Method	Closed loop plant+reference+relay; inject -75% CRAC capacity; measure onset-to-trip.
Measured	trip_source=L2_residual:3.40, fallback_latency_s=6
Budget	max_latency_s=15
Annex IV	Annex IV §2(g),§3,§4,§5
Note	latency is fault-onset -> first TRIPPED command, control-cycle resolution

CHK-DEADMAN · Dead-man revert on lost controller **PASS**

Requirement	If the relay goes silent beyond the dead-man horizon, the plant must autonomously revert to the safe always-on baseline without any command.
Method	Apply an accepted command; age it past the horizon; assert the plant reverts to baseline within one cycle.
Measured	command_actuated=True, reverted_to_baseline=True, deadman_horizon_s=90.0, revert_cycles=1
Budget	max_revert_cycles=2
Annex IV	Annex IV §3,§5

CHK-FENCE-ZOMBIE · Zombie fencing across failover (no dual-write) **PASS**

Requirement	After failover, a resurrected ex-leader must not be able to write to the actuator; the single-writer invariant must hold by fencing, not by agreement.
Method	Leader A writes; A dies; standby B takes over (epoch bump); zombie A re-publishes at its stale epoch; assert the actuator applied only A then B, never the zombie.
Measured	applied_under_A=0.5, applied_under_B=0.7, takeover_epoch=2, applied_after_zombie=0.7, dual_write=False
Budget	—
Annex IV	Annex IV §2(c),§3,§5
Note	enforcement is at plant-sim (the actuator), mirroring the dead-man philosophy

CHK-FENCE-RESTART · Restart re-acquisition (fresh sequence accepted) **PASS**

Requirement	A relay restart with a fresh in-memory sequence counter must be accepted by the actuator (epoch bumps per leadership session) and not rejected as a stale replay.
Method	Leader publishes seq 1..3; restart (same id) re-acquires; publishes seq 1; assert the epoch bumped and the fresh command is applied.
Measured	epoch_before=1, epoch_after_restart=2, is_leader=True, applied_flow=0.77
Budget	—
Annex IV	Annex IV §2(c),§3

CHK-HEALTH-LEAD · Coil-health early warning precedes safety trip **PASS**

Requirement	Slow coil fouling must raise a maintenance ALARM (advisory, not a trip) while the relay is still ARMED with airflow headroom — ahead of any acute trip.
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Method	Closed loop with MPC; inject slow fouling; assert ALARM precedes any L2 trip by the minimum lead.
Measured	alarm_tick=375, trip_tick=397, lead_s=22
Budget	min_lead_s=5
Annex IV	Annex IV §3,§9 (post-market monitoring)
Note	accelerated fouling rate for test; real lead scales ~1/rate (days/weeks)

CHK-HEALTH-LOADSHIFT · Coil-health load-shift false-alarm immunity **PASS**

Requirement	A change of operating point (load) must not be mistaken for coil degradation; baselines are per-load-bin.
Method	Commission one load bin; shift to a different bin with a higher structural residual; assert no alarm.
Measured	alarm_after_commission=False, alarm_after_load_shift=False
Budget	—
Annex IV	Annex IV §3,§4,§9